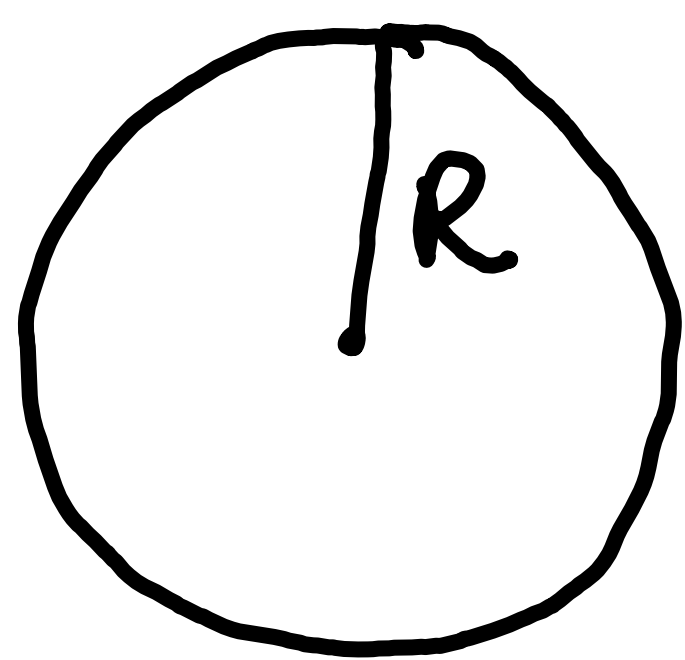


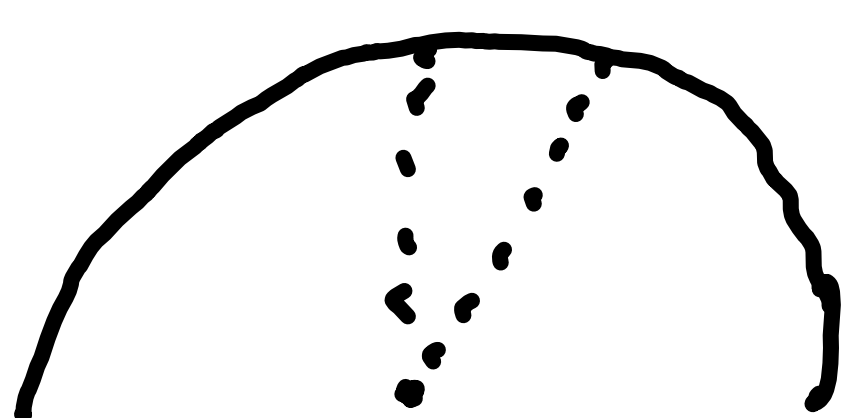
Monge gauge

Consider a circle with radius R .



the equation of the circle is

$x^2 + z^2 = R^2$, or in the upper hemisphere:



$$z = \sqrt{R^2 - x^2}$$

$$z_x = \frac{dz}{dx} = \frac{-x}{\sqrt{R^2 - x^2}}$$

$$z_{xx} = \frac{-1}{\sqrt{R^2 - x^2}} - \frac{x}{(R^2 - x^2)^{3/2}} \cdot (-2x)$$

$$= \frac{-1}{\sqrt{R^2 - x^2}} \left(-1 + \frac{2x^2}{R^2 - x^2} \right)$$

$$= \frac{-1}{R} \underbrace{\frac{1}{\sqrt{1 - (x/R)^2}} \left(-1 + \frac{2}{(R/x)^2 - 1} \right)}_{+1 \text{ for } \frac{x}{R} \ll 1}$$

$$\text{so } z_{xx} = -\frac{1}{R} + O\left(\frac{x}{R}\right)$$

↑
curvature!!

As long as $\frac{x}{R} \ll 1$, we can use the "Monge gauge" and approximate

$$H = \frac{z_{xx} + z_{yy}}{2} \quad \left. \vphantom{\frac{z_{xx} + z_{yy}}{2}} \right\} \text{2D}$$